PrivacyGuard Installation & Administrator Guide

Nov 9, 2026 · Author: Thangadurai Thambiannan

About this guide

This guide takes a Salesforce administrator from installing PrivacyGuard to running and monitoring masking jobs. PrivacyGuard is a Lightning app that finds potentially sensitive fields, masks them in sandboxes, and records every operation for review.

Item	Value
Product version	PrivacyGuard 0.5.0
Package version ID	04tQE00000jpLenYAE
Namespace	`privacyguard`
Audience	Salesforce administrators and approved PrivacyGuard operators
Support	privacyguard@sandboxshield.com.au

Before you start

- **Use a sandbox.** Masking, file deletion and trigger changes run only in sandbox orgs. PrivacyGuard refuses them in production, and there is no override.
- **Plan your recovery.** Masking and deletion can't be undone automatically. Make sure you can refresh the sandbox if needed.
- **Start small.** Test with a few synthetic records before running larger jobs.
- **Check your automation.** Validation rules, Flows and triggers can block masked records from saving. Review them before your first run (see sections 3 and 7).

Who needs which permission set

Permission set	Assign to	Lets them
PrivacyGuard Admin	PrivacyGuard administrators	Use discovery, configuration, masking, file deletion and the Audit Centre

Permission set	Assign to	Lets them
PrivacyGuard Operator	Users who run masking jobs	Create, preview, run and schedule jobs
PrivacyGuard Purge Operator	Users who delete files	Delete selected files
PrivacyGuard Tooling Access	Approved trigger administrators	Authenticate PrivacyGuard's trigger connection
PrivacyGuard Metadata Deactivation Break Glass	Approved trigger administrators	Deactivate and restore eligible Apex triggers

Field access for masking. PrivacyGuard masks a field only if the person running the job can edit it. Give operators, and anyone who schedules jobs, edit access to every field you plan to mask and to each object's `Data_Masking_Complete__c` checkbox. Use a permission set of your own for this, because PrivacyGuard's permission sets can't grant access to your custom fields. Records with a field the user can't edit fail and stay unmasked.

The **PrivacyGuard Retired Control A** and **PrivacyGuard Retired Control B** permission sets are kept only for upgrade compatibility. They grant nothing. Don't assign them.

1. Install and grant access

Installation takes a few minutes and changes no data. PrivacyGuard does nothing until you assign permissions and configure it.

1. Open the installation link supplied by PrivacyGuard and sign in to the sandbox where you'll use it. Select **Install for Admins Only**.
2. In **Setup › Installed Packages**, confirm that PrivacyGuard is listed with version 0.5.0.
3. Assign **PrivacyGuard Admin** to yourself and any other PrivacyGuard administrators. Assign the other permission sets only to users who need them.
4. Open **App Launcher › PrivacyGuard** and check that the Overview dashboard loads.
5. For each object you plan to mask other than Account, Contact and Lead, create a checkbox field named `Data_Masking_Complete__c`, default unchecked. Give operators edit access to it. PrivacyGuard ticks it on each record it masks successfully. Account, Contact and Lead already have this field from the package.

The dashboard shows how much of your accepted scope is configured, which objects and fields are active, and the status of the latest masking run.

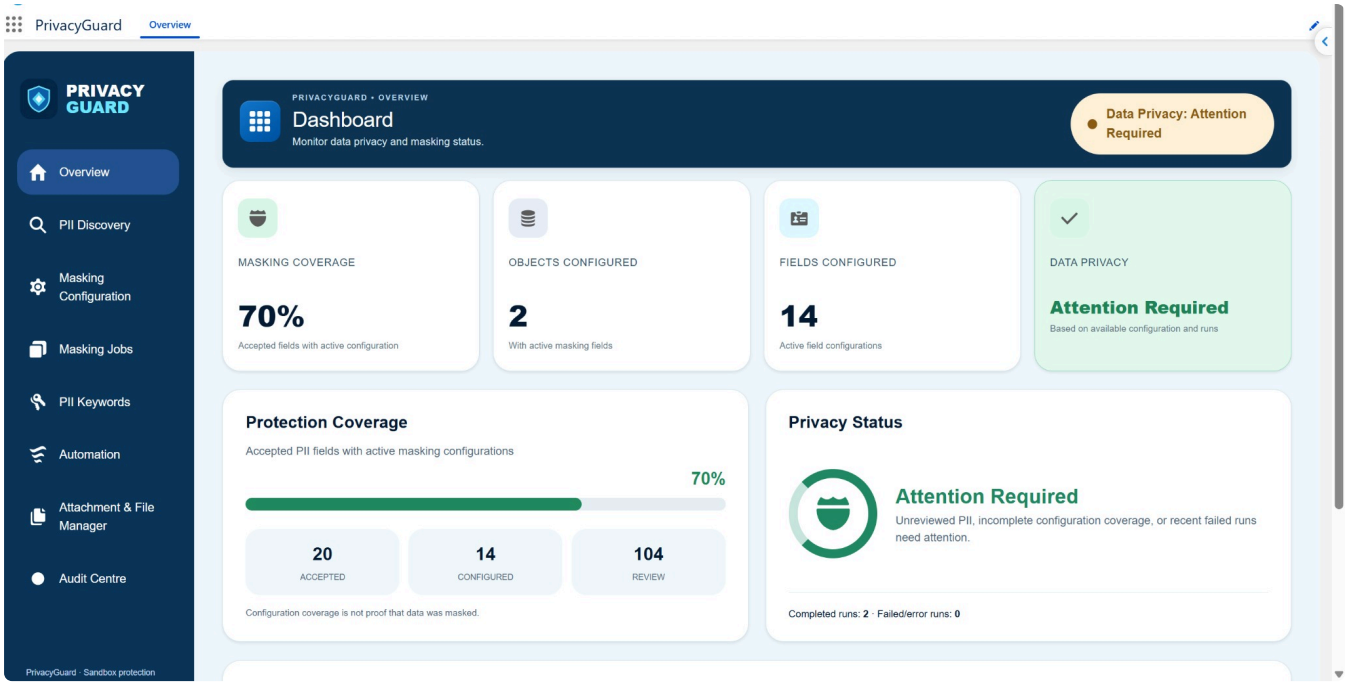


Figure 1. Dashboard: configuration coverage and privacy status. Coverage measures configuration only; it doesn't prove records have been masked.

2. Set up Apex trigger management (optional)

Do this setup only if you want PrivacyGuard to deactivate and restore unmanaged Apex triggers. Discovery, configuration, masking and file management work without it. PrivacyGuard connects only to the same sandbox it's installed in, and each approved operator signs in with their own account.

2.1 Configure OAuth

1. In **Setup › Named Credentials › External Auth Identity Providers**, open **PrivacyGuard Tooling OAuth** and copy its callback URL.
2. In the same sandbox, create or update a Salesforce OAuth app (an External Client App or Connected App). Use the exact callback URL, enable the **api** and **refresh_token** scopes, and keep **PKCE** enabled.
3. Back in **PrivacyGuard Tooling OAuth**, enter the app's client ID and client secret, plus your sandbox's authorization and token endpoint URLs. Don't share the secret.
4. Open the External Credential **PrivacyGuard Tooling OAuth**. Confirm the scope is `api refresh_token` and that the principal **PrivacyGuard_User** uses the **Per User Principal** identity type.

2.2 Set the Named Credential

Edit the installed **PrivacyGuard Tooling** Named Credential. Don't create a new one.

Setting	Value
API name	`privacyguard__PrivacyGuard_Tooling` (installed)
URL	This sandbox's own My Domain URL, for example `https://yourcompany--uat.sandbox.my.salesforce.com`
External Credential	PrivacyGuard Tooling OAuth (installed)
Enabled for Callouts	On
Generate Authorization Header	Checked
Allowed Namespaces for Callouts	Leave blank

The URL must be this sandbox's own My Domain. Before every operation, PrivacyGuard checks that the connected org has the same Org ID and is a sandbox, and refuses otherwise.

2.3 Authorize each operator

1. Assign **PrivacyGuard Tooling Access** and **PrivacyGuard Metadata Deactivation Break Glass** to approved trigger administrators, in addition to their usual PrivacyGuard access. They also need the Salesforce permissions to use the API and change Apex.
2. Each operator opens their personal **Settings › External Credentials** (or **Authentication Settings for External Systems**), selects **PrivacyGuard Tooling**, allows access, and signs in to this sandbox. The status should show **Configured**.
3. Test with one eligible trigger: deactivate it and check it shows Inactive, then restore it and check it shows Active.

After a sandbox refresh or My Domain change, recheck the Named Credential URL, the callback URL and the OAuth app, then ask each operator to sign in again.

3. Find your way around

PrivacyGuard opens on **Overview**. Its left-hand menu lists every area of the app, in roughly the order you'll use them.

Menu item	Use it to	Needs
Overview	See masking coverage, configured objects and fields, privacy status and run counts	Admin
PII Discovery	Scan for candidate sensitive fields and accept or remove them	Admin
Masking Configuration	Choose a strategy for each accepted field and activate it	Admin
Masking Jobs	Build jobs, preview, run, schedule, cancel and rerun	Admin or Operator
PII Keywords	Add or remove discovery keywords	Admin
Automation	Deactivate and restore eligible Apex triggers	Trigger setup (section 2)
Attachment & File Manager	Inventory, export and delete files and attachments	Admin or Purge Operator
Audit Centre	Review every operation and configuration change	Admin

Automation during a masking run

When a job runs, PrivacyGuard turns on bypass flags in the `Automation_Bypass__c` custom setting and turns them off again when the run finishes. Your own Flows, validation rules and triggers pause only if they check these flags. Ask your developers to add a condition such as `$Setup.privacyguard__Automation_Bypass__c.Bypass_Validation_Rules__c = false` to any automation that should skip masked updates. Alternatively, deactivate eligible Apex triggers in the Automation tab (section 7).

4. Discover and review sensitive fields

Discovery suggests fields that may hold personal data. You decide what to mask. It reads field definitions only, never record values, and is safe to run in any org.

4.1 Run a scan

1. Open **PII Discovery** and select **Run New Scan**. The scan checks every object you can query and update, one object at a time. The progress bar shows objects scanned, candidates found and errors.
2. When the status reads **PII Discovery: Completed**, read any yellow warning. A classification lookup warning means Salesforce Data Classification couldn't be read for some objects; keyword and field-type checks still ran. Details are in the Audit Centre.
3. Use the search box and the status filter (Pending, Accepted, Removed) to work through candidates. Expand an object to see each field, its type and why it was flagged.
4. Select fields and choose **Accept Selected** or **Remove Selected**, or act on a whole object with **Accept Object** or **Remove Object**.

A field is flagged when any of these apply:

- Its type is Email or Phone, or it's an encrypted field.
- Its label or API name contains a PII keyword as a whole word (see 4.2).
- Salesforce Data Classification gives it a compliance group, or a sensitivity of Confidential, Restricted or Mission Critical.

Accepting a field doesn't mask it. It creates an inactive rule in **Masking Configuration**, where you choose the strategy (section 5). Fields that already have a rule aren't suggested again. Discovery can't guarantee every sensitive field is found, so review your custom fields too.

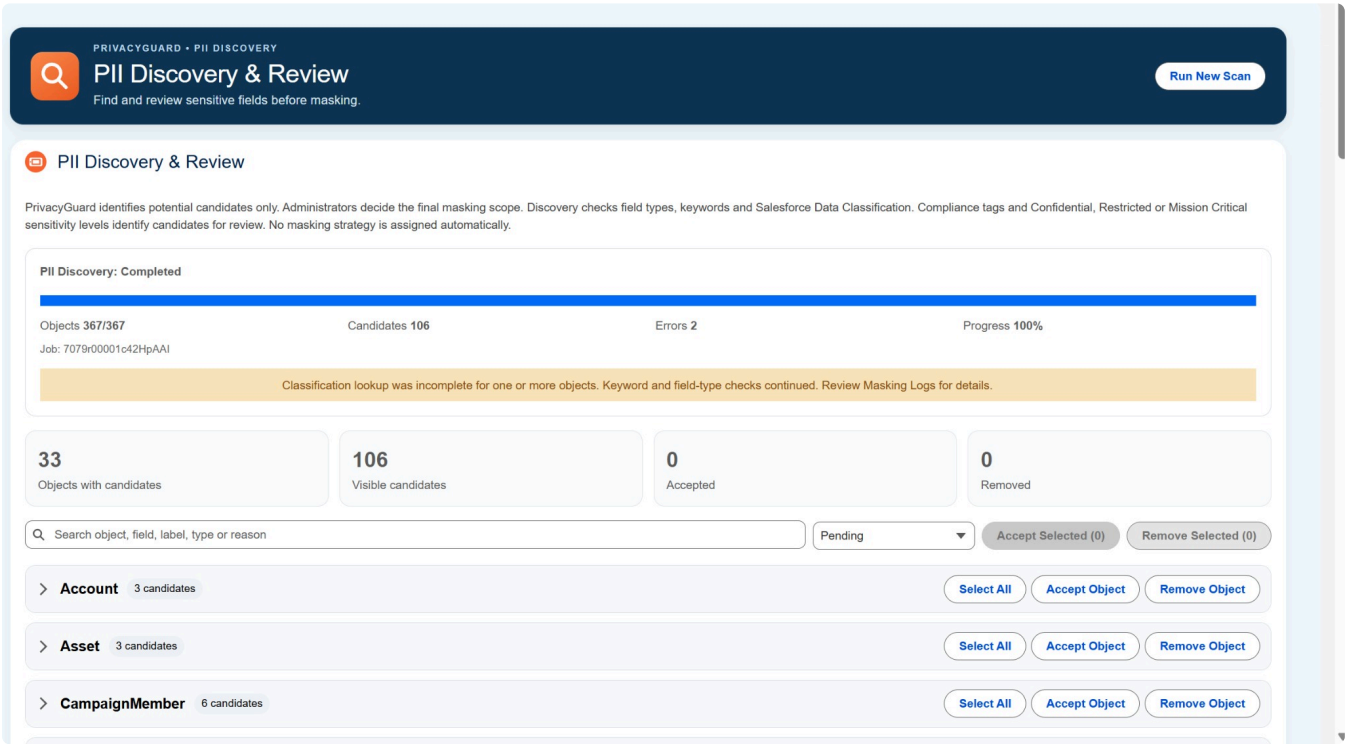


Figure 2. PII Discovery & Review: scan progress, candidate counts and per-object review. This example shows a classification lookup warning.

4.2 Maintain PII keywords

Open **PII Keywords** to tune discovery for your organisation. PrivacyGuard ships 31 default keywords, such as address, bank_account, date_of_birth, passport and ssn.

- To add one, enter it under **New keyword**, add optional **Notes**, and select **Add Keyword**. New keywords are active immediately.
- To stop using one, including a PrivacyGuard default, select **Delete**.

Changes apply on the next scan. Matching ignores case and treats underscores, hyphens and camelCase as word breaks, so `date_of_birth` matches a field labelled "Date of Birth".

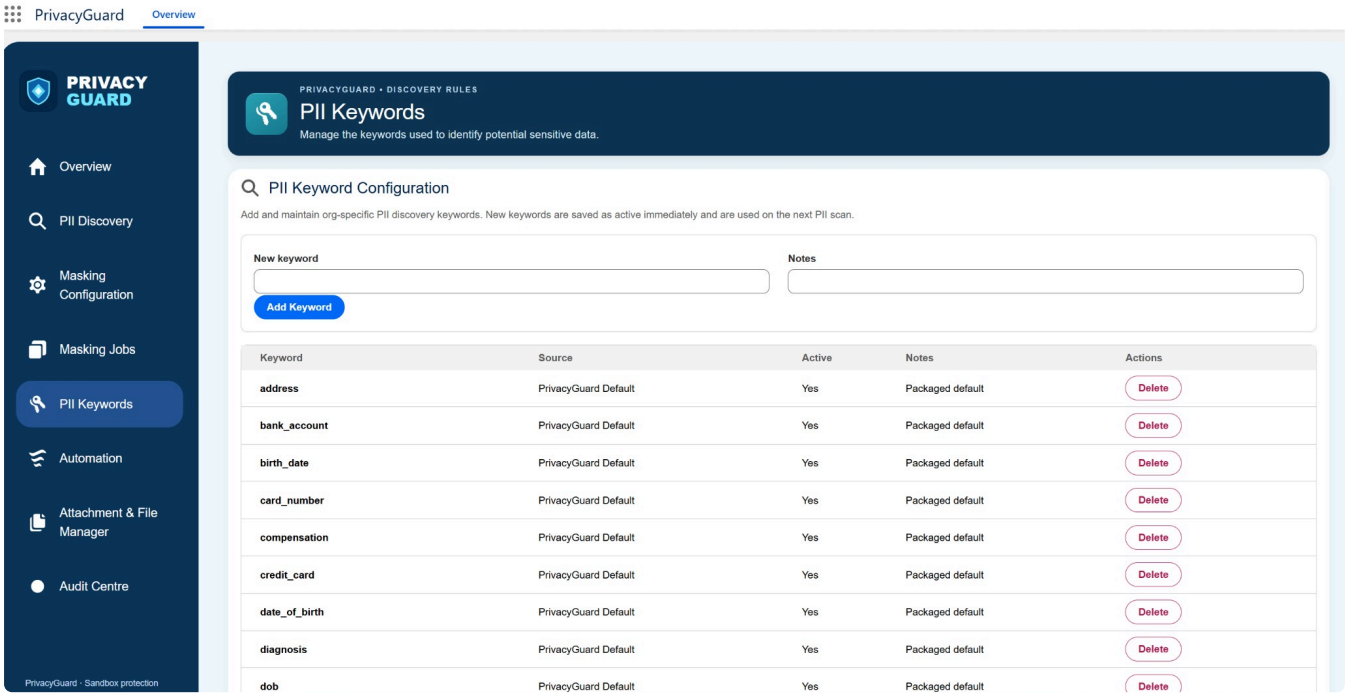


Figure 3. PII Keywords: packaged defaults and organisation-specific discovery terms.

5. Configure field masking rules

A field is masked only when it has an active rule with a strategy. Rules apply to every job that includes the field's object.

1. Open **Masking Configuration**. Fields you accepted in discovery appear here, grouped by object, with Source **PII Discovery**. Use **Add Fields** to add any other field.
2. For each field, pick a **Masking strategy**. The list shows only strategies valid for that field's type.
3. Fill in **Pattern / value** if the strategy needs one (see the table below).

4. Turn on **Active**. The **Need strategy** count at the top should reach 0 before you run a job.
5. Use the delete icon to remove a rule. The field can then be found by discovery again.

What it does	Pattern / value
Random text of a set length	Optional length (default 8)
Random digits, keeping the phone format	None
Random email at the original domain	None
Obviously fake email: `sandbox.<code>@invalid.com`	None
Random name from a built-in list of 10	None
Random value from your own list	Values separated by `|`, e.g. `Gold|Silver|Bronze` (255 characters max)
Random number in a range	`min|max|decimal places`, e.g. `100|999|0`
The same fixed value on every record	The value to write
Regular-expression replacement	`expression=>replacement`
Clear the field	None; not allowed on required fields

If an email domain could identify your organisation or customers, use the fake-email strategy rather than random email. Check that your validation rules accept the masked values, especially for phone, email and picklist-dependent fields.

Completion checkbox

Every object you mask needs an editable checkbox, normally `Data_Masking_Complete__c`, defaulting to unchecked. PrivacyGuard ticks it only when every configured field on a record was masked successfully. Jobs set to **Unmasked records only** use it to skip finished records. Account, Contact and Lead have it from the package; create it on any other object (section 1). It can't also be a masking target.

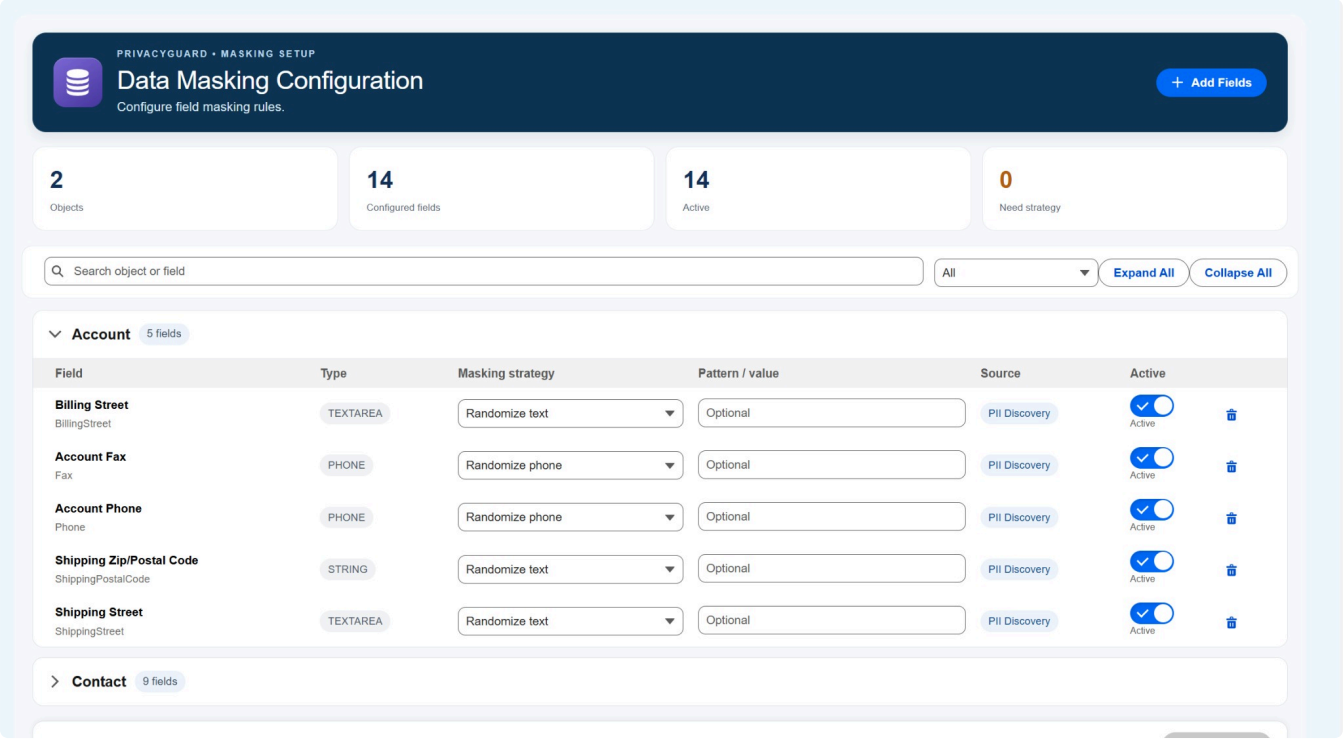


Figure 4. Data Masking Configuration: field rules, strategies, patterns and active settings.

6. Create and run masking jobs

A job groups configured objects into one reusable, controlled run. Jobs run only in sandboxes. Masking can't be undone, so always preview and test with a small job first.

6.1 Create a job

1. Open **Masking Jobs** and select **Create Masking Job**.
2. Name the job and add a description.
3. Select the objects to include. Only objects with active rules are offered. Set their order if the job runs serially.
4. Choose the options:

Option	Choices
Mode	**Full** masks every record. **Unmasked records only** masks records whose completion checkbox is unticked.
Execution	**Serial** runs objects one after another; if one fails, the rest are skipped. **Parallel** runs all objects at once.

Option	Choices
Batch size	Records per batch, 1–2,000. Default 200. Lower it if automation or validation errors appear.

6.2 Preview and run

1. From the job's **Actions** menu, open **Preview**. It shows before and after values for up to 5 records without saving anything.
2. Select **Run**. Review the scope, acknowledge the safety warning, and start.
3. Watch **Status** and open the run details. Each object shows records processed and failed.

Before starting, PrivacyGuard checks that the org is a sandbox, the job is active and not already running, each object has active rules, and each completion checkbox exists and is editable. If anything fails, nothing starts and you'll see the reason.

Status	Meaning
Running	Batches are in progress
Completed	All objects finished with no failed records
Completed with Errors	Some records or objects failed; others were masked
Cancelled	Stopped by a user; batches already committed stay masked
Failed	The run couldn't complete

A record is saved only if every configured field masked successfully, so no record is left half-masked. Failed records stay unticked and are listed in the run details and the Audit Centre.

6.3 Fix failures and rerun

1. Open the run details. Each failure lists the record ID, object and field, but not the Salesforce error text, which PrivacyGuard doesn't store. To see the actual error, open the record and try the same edit yourself. The most common causes are missing edit access to a masked field or the completion checkbox, validation rules, required fields, and automation that rejects the masked values.
2. Fix the cause in your configuration or automation.

3. Choose **Rerun** on a run marked Completed with Errors or Failed. Only the failed records are processed again.

6.4 Schedule a job

After a successful manual run, you can schedule the job daily, weekly on a chosen day, or monthly on day 1–28 or the last day. Schedules run in the scheduling user's time zone, with that user's permissions. Check **Next Scheduled** after saving. Cancel a schedule from the job's **Actions** menu.

6.5 Cancel a run

Cancel stops any batches that haven't started yet. Records already masked stay masked and can't be restored.

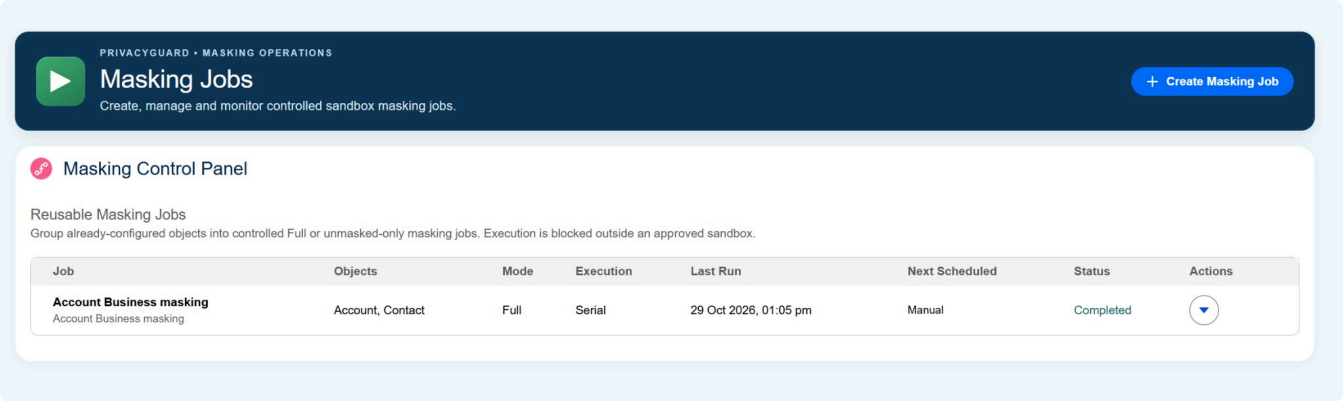


Figure 5. Masking Jobs: reusable job configuration, schedule and latest status.

7. Manage Apex triggers

Use **Automation** to switch off unmanaged Apex triggers that would block masking, then switch them back on afterwards. It needs the setup in section 2, and works only in sandboxes. Flows and validation rules can't be changed here; use the bypass flags in section 3 for those.

1. Open **Automation** and select **Scan Triggers**. The counters show triggers discovered, currently active, tracked for restore and selected.
2. Search or filter by state. Each row shows the object, trigger name, current state and whether PrivacyGuard can change it. Namespaced (managed package) triggers and PrivacyGuard's own audit triggers are read-only.
3. Select the eligible triggers and choose **Deactivate Selected**. PrivacyGuard records each trigger's original state before changing it.

4. Wait for the **Deployment** column to confirm the change. It shows success only after reading the trigger's actual state. *Queued* or *Check required* means the change isn't confirmed yet. Use the status check to try again.
5. Run your masking job.
6. Select the same triggers and choose **Restore Selected**. Only triggers PrivacyGuard deactivated can be restored. Confirm each one shows **Active** again.

Always restore and verify every trigger you changed before handing the sandbox to its users.

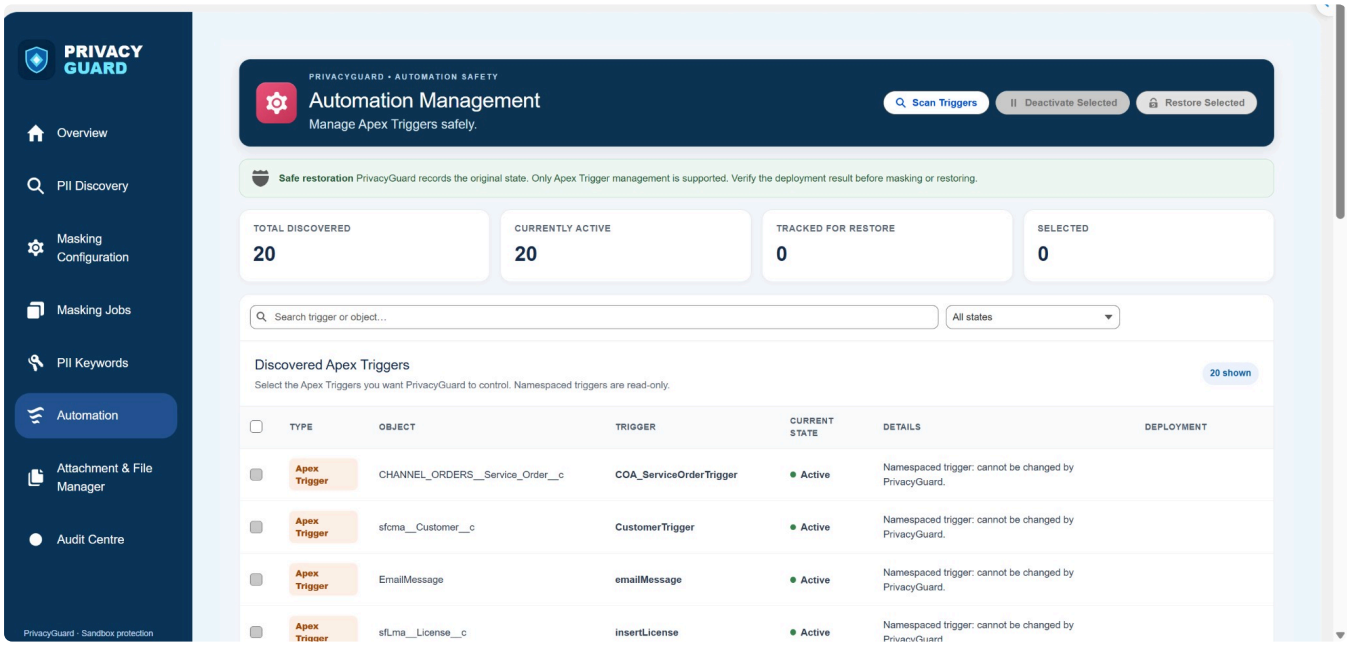


Figure 6. Automation Management: discovered triggers, current states, and deactivate and restore actions. Namespaced triggers can't be changed.

8. Review and remove attachments and files

Attachment & File Manager helps you find and remove Salesforce Files and legacy Attachments that sandbox users shouldn't see. You delete files by selecting them; there is no bulk purge. Deletion works only in sandboxes, needs the **Purge Files** permission, and can't be undone.

8.1 All files: inventory and selected deletion

1. Select **Scan All Accessible Files**. PrivacyGuard lists the files and attachments you can see, 100 per page, with titles, owners, sizes and linked records. Totals show Salesforce Files, legacy Attachments and total bytes. Use **Stop scan** to end early.
2. Select **Download CSV report** to keep a record or share it for review.

3. Select the items to delete, up to 100 at a time, and type **DELETE** to confirm. Deletion uses your own Salesforce permissions, so you can delete only what you could delete in standard Salesforce.

The inventory respects your permissions and file sharing. Other users' private files may not appear, and a file with no visible record link may still be in use elsewhere.

8.2 By object: counts only

Choose up to 10 objects and scan them to see how many attachments and file links their records have, counting only what you can see. Use this to find where files are concentrated, then delete them from **All files**.

Object-wide purge was removed in version 0.3.0. The **Object-wide purge disabled** button can't be used, and any purge scheduled under an earlier version is refused without deleting anything. Remember that deleting a Salesforce File removes all its versions and every link to it, including links to records on other objects. PrivacyGuard doesn't check what's inside files.

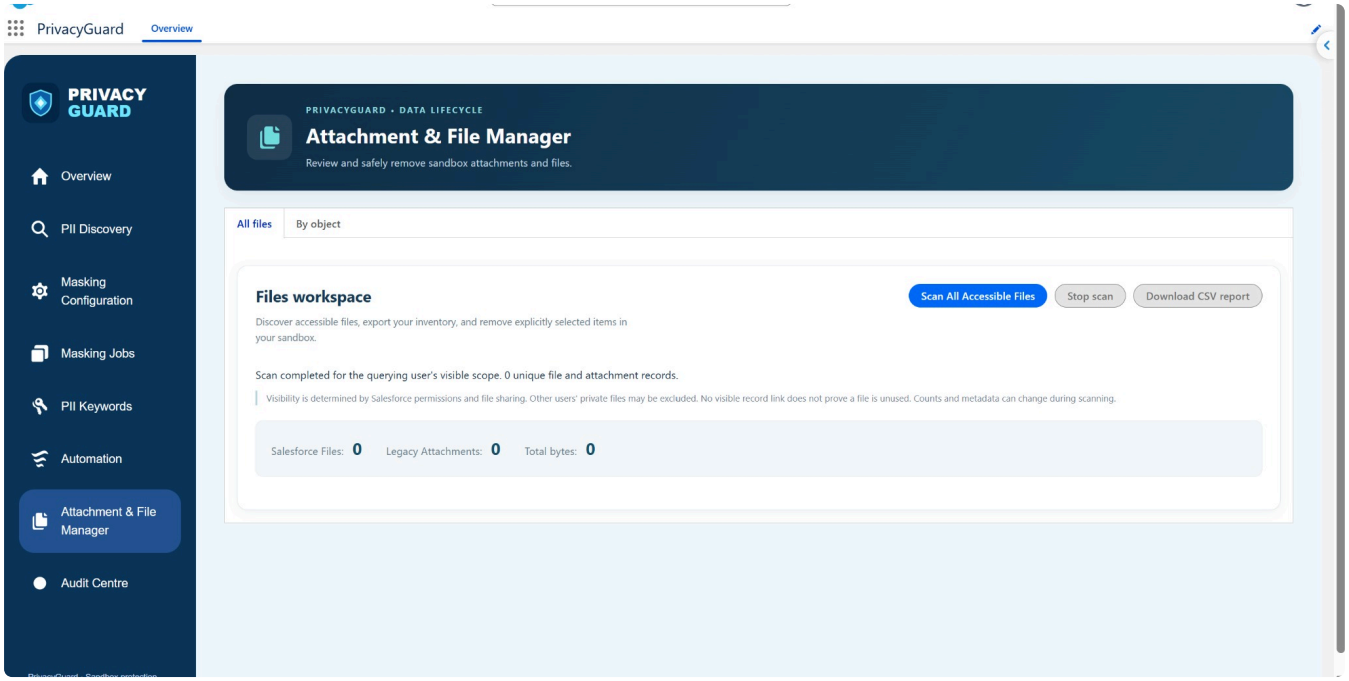


Figure 7. Attachment & File Manager: file inventory, scan controls and CSV export. This example sandbox has no visible files.

9. Review activity in the Audit Centre

The **Audit Centre** records every scan, masking run, failure, deletion, trigger change and configuration change. Use it to investigate a problem or show what was done.

1. Set the filters: **Category** (Data Masking, PII Discovery, Automation, Files, Configuration, Jobs), **Status**, **From** and **To** dates, **User name**, and **Job, object, field or record ID**.
2. Select **Refresh / Apply**. The counters show activity, failed entries and configuration or job changes loaded.
3. Select **Details** on any row to see the message, suggested action, and the record, object and field involved. Configuration changes show before and after values, with secrets redacted.
4. Select **Export loaded rows** to download what's on screen. Counts and exports cover loaded rows only, so use **Load more** to include older activity.

Dates use your Salesforce time zone. Audit entries are permanent: nobody, including administrators, can edit or delete them. Failure entries show a standard message instead of the original Salesforce error, so no field values are kept in the log. Audit Centre access still needs **View PrivacyGuard Audit** or the Admin permission set.

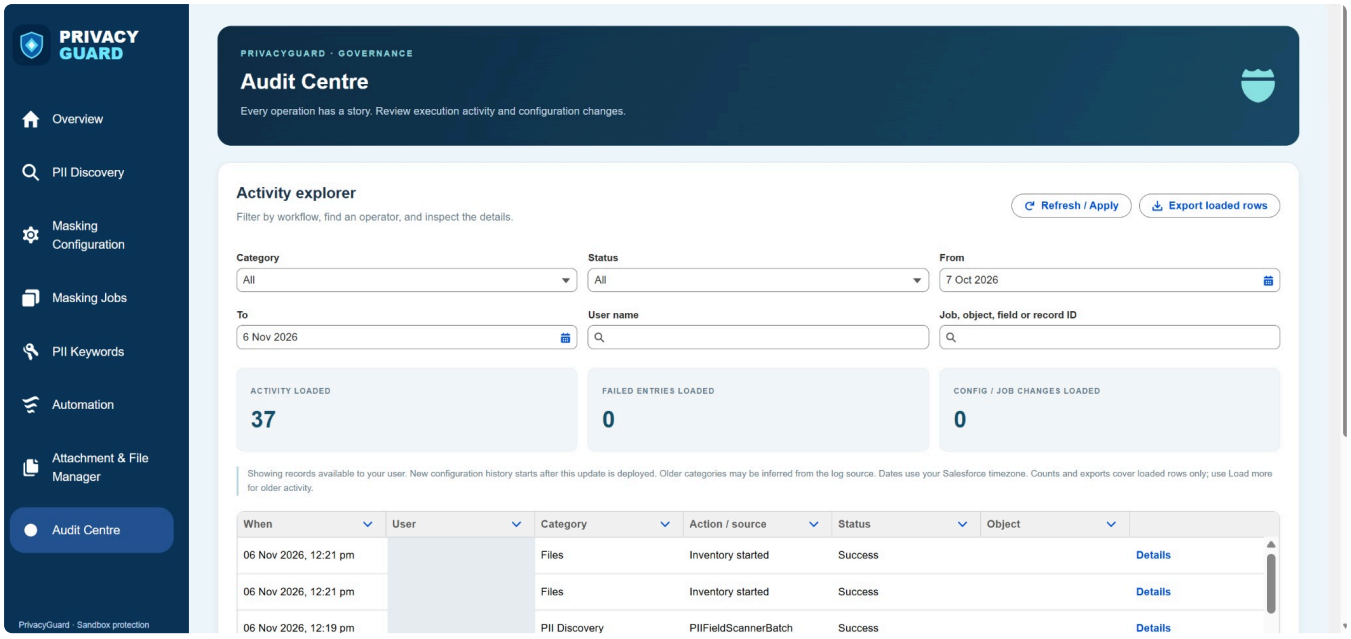


Figure 8. Audit Centre: activity filters, summary counters and log entries. Personal names are hidden in this image.

10. Routine checks, troubleshooting and help

Routine checks

- [ ] After each run: check the job status, rerun failed records, and restore any triggers you changed.
- [ ] Before an upgrade or sandbox refresh: let active runs finish or cancel them, and review schedules.

- [ ] After an upgrade: check permission set assignments and configuration, then do a small test run.
- [ ] After a refresh: recheck completion checkboxes, the Named Credential URL, OAuth settings and schedules. Ask trigger operators to sign in again.
- [ ] Regularly: review who has PrivacyGuard permissions, keep Audit Centre access restricted. Audit entries can't be deleted, so keep an eye on data storage; they're cleared when the sandbox is refreshed.

Troubleshooting

You see	Likely cause	What to do
"Execution is blocked" or a sandbox-only message	The org isn't a sandbox	Masking, deletion and trigger changes only run in sandboxes
A permission error	A PrivacyGuard permission set is missing	Assign the permission set for that task (see About this guide)
Job won't start: missing configuration or checkbox	An object has no active rules, or no editable completion checkbox	Activate rules in Masking Configuration; create or grant edit on `Data_Masking_Complete__c`
Job won't start: already running	A previous run is still active	Wait for it to finish, or cancel it
Completed with Errors	The running user can't edit a masked field, or validation rules, required fields or automation rejected some records	Read the run details, fix the cause, then **Rerun**
Classification lookup warning in discovery	Data Classification couldn't be read for some objects	Keyword and field-type checks still ran. Review the warning in the Audit Centre.
Trigger change shows Check required	The deployment hasn't been confirmed yet	Run the status check again before masking or restoring
Trigger connection fails	OAuth, Named Credential URL or operator sign-in issue	Recheck section 2; after a refresh, sign in again

Get help

Email **privacyguard@sandboxshield.com.au** with:

- the installed package version (Setup › Installed Packages)
- what you were doing, and the job or run ID
- the error text from the screen or the Audit Centre

Remove sensitive data, passwords, client secrets and tokens from screenshots and logs before sending them.